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1.0 Revision History

REVISION	DATE	NATURE OF CHANGES
Initial	February 28th, 2018	Initial release. Obsoletes the following OPMP's: 1.1, 1.2, 1.4, 1.5, 1.6, 2.1, 2.2, 2.3, 3.2, 3.4, 3.5, 3.7 and includes their intent.

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2.0 Export Control Requirements

To ensure compliance with current export control laws and regulations of the Canadian and U.S. governmental agencies regarding the export or transfer by any means, directly or indirectly, of any controlled goods or technical data, in accordance with MMP375, the reader shall refer to OPMP 7.1, OPM Compliance to Applicable Export Control Regulations.

In the event any data is received, created, handled or transferred during the execution of this procedure, it is the employees' responsibility to ensure all export control requirements are addressed, all data has been classified, and anyone with whom they are sharing the data, has the appropriate clearances.

In the event any clarifications or questions arise related to export control, the reader is requested to consult their departmental International Trade Compliance focal point.

3.0 Scope

The scope of this procedure is to describe Operations Program Management (OPM) role in achieving a successful industrialization of a new program. OPM is responsible for Production readiness process for the transition of components, engines (including APUs) and powerplants from proposal through serial production (MMP 219, MMP 220). OPM strives for each program to deliver a producible product that is sourced to meet program objectives of cost, yield, quality, schedule and Change Order effectivity.

As soon as an MPI request is raised by Marketing to study a new program, OPM takes an active role in defining: NPI Operations Budget, the Program sourcing strategy and risks, the Level 2 Operations plan and the Operations PDRS as explained in OPMP 3.10 Industrialization - Pre-P0 up to P2. Once the Product Proposal phase is completed, the Definition, Validation and Delivery phases start, typically covering activities from Passport 2 - Permission to Continue Detail Design to Passport 4 - Permission to Deliver Serial Production. This procedure defines the roles and responsibilities of OPM during these industrialization execution phases.

4.0 Acronyms

AFS	Assembly Floor Sheet
APU	Auxiliary Power Unit
AQP	Airworthiness Quality Plan
A&T	Assembly and Test
BS	Build Specification

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CAR	Capital Acquisition Request
CBN	Cost Basis Note
CBOM	Concept Bill Of Material
CIFER	Configuration Identification For Experimental Requirements
CMM	Configuration Management Manual
CO	Change Order (formally EC)
CR	Change Request (formally eRFC)
DBOM	Development Bill Of Material
DID	Design Intent Document (formally Interim Design Release)
DP	Design Project (formally Design Job)
DPM	Digital Product Management
DR	Discrepancy Report
DTIS	Development Test Information System
EBOM	Engineering Bill Of Material
EIS	Entry Into Service
EMO	Engineering Material Order
EMWO	Engineering Material Work Order
ESA	Engineering Source Approval
ESK	Engineering Sketch
FAA	Federal Aviation Administration (USA)
FSC	Factory Standard Cost
FSR	Field Service Representative
I.O.	Internal Order
IPEC	Intellectual Property and Export Control
IPPS	Integrated Power Plant System

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IPT	Integrated Product Team
ITC	International Trade Compliance
MA	Manufacturing Approval
MPI	Market Proposal Inquiry
NPI	New Product Introduction
OEM	Original Equipment Manufacturer
OEUP	Original Equipment Uprade Plan
OPE / OPA	Operations Program Engineer / Operations Program Analyst
OPM	Operations Program Management
OPMP	Operations Program Management Procedure
OPMWI	OPM Work Instruction
POA	Product Organizational Approval
PCOR	Program Customer Operations Revie
PFMEA	Process Failure Mode Effect Analysis
POC	Parts of Concern
PLM	Product Lifecycle Management
PMT	Product Management Team
QRA-A	Quarantine Release Authorisation for Assembly
RT	Engineering Request for Test
SFV	Strategic Financial Valuation
TATP	Temporary Assembly and Test Procedure
UPPAP	United Technologies Production Part Approval Process
XSK	Experimental Sketch

5.0 Definitions

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Consolidated Report: Monthly report used to monitor the OPM capital and managed expenses for NPI and Legacy Programs. It shows the yearly budget and actuals by month.

Engineering Red Flag Notice (RFN): Initiated and issued by Engineering, with the concurrence of OPM, to authorize action to stop procurement of hardware (raw material, part assemblies, detail, etc.) or engine manufacture or shipment (P&WC 7736). It contains commit dates from Engineering, Design, Drafting and DPM. Publication of a CO closes the RFN.

Legacy Program: A program that has reached aircraft entry into service.

Investment Audit: Regular financial exercise that compares the latest estimates of projected spending, cash flows and other program benefits against the original assumptions set forth in the capital appropriation request.

Key Component: Is a part unique to the program which has one or more of the following characteristics: high complexity, high cost, long lead time, new supplier, new technology, dual sourced, etc (ref. complete list of characteristics in OPMP 5.10 Change Order Management).

Limit Notice (LN): Issued by Engineering, with the concurrence of OPM, to authorize action to limit production of raw material, parts, assemblies or end items (P&WC 7741).

Manufacturing Approval: Is First Article Inspection process for Authorized Manufacturing Centre or Engine Centres which demonstrates that the production processes, documentation, tooling and personnel are capable of producing engines that meet production requirements for Transport Canada.

New Product Introduction Program: An engine program that has not reached Aircraft Certification or is prior to Passport 4 - Permission to Deliver Serial Production and has no engines delivered to an Operator or first end-user.

NPI Operations Budget: Operations budget established and reviewed by OPM at every Passport and Investment audit. Please refer to OPMP 3.10 for more details.

OEM Consignment Stock: Hardware that P&WC holds under its inventory, but which is physically at the OEM. It is used to allow quick replacement on site in the event of a part failure.

Parts of Concern (POC): Manufacturing Analyst will identify POC based on part complexity, ESA requirements, past QN history and MFA from Project Engineering.

Pre Production Engine: Also known as X-engines or Prototype Engines, an engine that has been assembled, tested and received an authorized release certificate as “prototype” status on a TCCA Form One by a P&WC Engine Centre Production environment and intended to be approved for in-service operation.

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Risk Hardware: Any engine part that must be ordered to support production engine build where the required definition release is not available within lead time of the part to meet a customer, quality or program commitment.

Shipment Restriction CO: Formerly called a "Pink EC", is a CO issued with pending Engineering substantiation or waiting customer concurrence or regulatory acceptance. It allows procurement of parts. It has an incorporation date, cancels the previous configuration and requests the new configuration in SAP.

6.0 References

EOP 22	Limit Notice/Red Flag Notice
EOP 103	Design Process and Design Reviews
EOP 257	Development Engineering Production Readiness Process
ECOP 07-03-02	CIFER, EEM, EPNS, XSK
MMP 219	Integrated Product Management Process
MMP 220	Passport Review Process
MMP 300	Work Transition
MMP 348	Sourcing
MMP 375	International Trade Compliance
MMP 501	New Product Introduction FSC Management
MMP 515	Engine Program Investment Guidelines (CAR's)
OPMP 3.10	Industrialization - Pre-Passport 0 up to Passport 2
OPMP 4.10	Development Hardware Management
OPMP 4.11	Production Hardware Management
OPMP 6.10	Management of Delivered Engines
OPMP 7.1	OPM Compliance to Applicable Export Control Regulations
OPMWI 3.11.1	Establish Component Strategy
OPMWI 3.11.2	Raise OPM Program CAR
OPMWI 3.11.3	Manage Capital Equipment

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OPMWI 3.11.5	Load Production Intent Build Specification in SAP
OPMWI 3.11.6	Manage Pre-Production Engines
OPMWI 3.11.7	Manage OEM Consignment Inventory
QMSP 10-02	Release Documentation Requirements
QMSP 10-06	Manufacturing Approval - Engines and Sub-Assemblies
QMSP 10-08	Airworthiness Quality Plans
P&WC 10367	Component Strategy Form
P&WC 11847	NPI Operations Budget Form
P&WC 11895	Program Sourcing Strategy Form

7.0 Program Operations Level 1 & 2 Plan

Prior to Passport 2, the OPE / OPA has defined the Program Operations Level 1 & 2 Plan that integrates all critical industrialization activities that need to be executed or followed up to achieve a successful entry into production. Throughout the different program phases, the OPE / OPA uses this plan as a tool to measure progress, ensures no key activities are missed resulting in significant delays or inefficiencies and initiates corrective actions whenever required. It can also be used to communicate industrialization status at PCOR, and other communication forums.

The Program Operations Level 1 & 2 Plan is a backbone to OPMPs 3.10 and 3.11 as it integrates each industrialization topic with respect to Passport gates.

Note: Please refer to OPMWI 3.10.4 Establish Program Operations Level 1 and 2 Plan for more details.

8.0 Delivery of Development Engines

Once a Market Proposal Inquiry (MPI) request is raised by Marketing to gather information required to support a new engine proposal, Advanced Design activities start. From Passport 0 - Permission to Study through Passport 3 - Permission to Order Production Hardware, Engineering transitions from Concept Design (P0-P1) to Pre-detail Design (P1-P2) and finally Detail Design (P2-P3) as described in EOP 103. OPM's involvement in development activities starts in the Concept phase and becomes more active in the Pre-detail Design phase with increasing intensity during the Detail Design phase.

8.1 IPTs and PMT

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The development of a new engine is typically divided into Modules (compressor, turbines, externals, etc.). Project Engineering raises multiple Design Projects (DP) to design and substantiate each engine component part of a module. Integrated Product Teams (IPTs) are used to execute and monitor the progress of DPs. The goal is to release a configuration to trigger procurement actions and allow substantiation activities.

As Operations' representative, the OPE / OPA takes an active role in various Engineering IPTs and in the Product Management Team (PMT). During the IPTs, OPM's mandate is to understand the Module definition progress and highlight any potential operational risks that could jeopardize development milestones and / or entry into production. In case of a derivative engine, the OPE / OPA can consult the PCOR and available quality reports of the base model to understand associated risks. At this early stage, involvement of Manufacturing Analyst is critical to assess producibility aspects. The OPE / OPA also supports any demand requiring Operations' feedback such as sourcing, lead times, capacity, etc.

During the PMT, the OPE / OPA provides a high level development material status. It typically includes: top pacer parts on order, pending Engineering Material Orders (EMOs) and definition release dates required for unreleased parts. The accuracy and timely availability of the EBOM / CBOM are essential for proper hardware management in DTIS. If these criteria are not met, the situation shall be escalated at the PMT to ensure we achieve an accurate hardware control and meet program milestones.

The PMT is the forum to communicate any concerns on the Program Development progress that can result in missing operational and customer commitments in terms of cost, schedule, producibility or quality. The OPE / OPA is responsible to communicate the overall program status provided in the PMT, to Operations' organizations (Canadian Manufacturing sites, Supply Management, Engine Centres, etc.) when deemed necessary.

Note: Refer to the MMP 219, Appendix H to obtain a more detailed list of key Operations deliverables expected at the IPT and PMT levels.

8.2 Key Development Engineering Activities

Development Engineering has the mandate of building and testing development and customer engines. They issue and maintain the program test plan (Development Level 2 Plan) that documents each test required to certify the program. They also load the EBOM / CBOM into DTIS and adjust the parts list as required to obtain the specific configuration of each development engine required.

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Development Engineering is responsible to raise EMOs for top assembly parts, create temporary experimental part numbers such as cifer, XSK, ESK (ref. to ECOP 07-03-02) and raise and follow up Engineering Material Work Orders (EMWOs) whenever required.

Per EOP 257, they are responsible for Development Production Readiness process. This is the foundation of the Assembly and Test (A&T) production readiness activities for which OPM is responsible, as described in Production Readiness Section 13. The involvement of the OPE / OPA is therefore essential to ensure that key activities such as Discrepancies Report (DR) closure plan, Development Process Failure Mode Effect Analysis (PFMEA), tooling and Assembly Floor Sheet (AFS) validation and test cell readiness, are progressing per plan during development phase.

8.3 Development Configuration Management

In order to accelerate the procurement cycle, advance release drawings are usually issued ahead of the DP closure. This allows OPM to initiate the hardware procurement activities as described in the following section. Any fit, form or function changes affecting released parts will result in a new part number or Mandatory Revision Letter (ref. CMM sections C7 and B9). Temporary part numbers can also be used to trigger procurement actions (ref. ECOP 07-03-02).

To facilitate hardware management, Engineering needs to produce Design Intent Documents (DID) on all DPs as early as possible in order to obtain a formal EBOM in Product Lifecycle Management (PLM). To minimize risk of delays, first release of EBOM shall be targetted between CDR and FER. As the design evolves, multiple versions of EBOM are generated in PLM.

8.4 Development Hardware Management

Development hardware management activities start with the identification by the PMT of a reference model similar to the new engine program. Based on this model, OPM can start the long lead times analysis to highlight the required drawings release dates for the Key Components. A Development kit can be defined and loaded in SAP to secure common parts especially for derivative programs. Risk mitigation strategies to procure long lead time raw material for development needs as well as production are established. This analysis gets more refined and detailed following the identification and release of all new unique parts. As soon as the drawings become available, the OPM Planner initiates the procurement process.

DTIS is the prime tool to be used to track the hardware status. OPM needs to follow up closely engine configuration changes and DTIS updates in order to ensure that the proper hardware is ordered (refer to OPMP 4.10 Development Hardware Management for details).

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In the situation where the procurement process is started too late following delays in drawings' release, it is the responsibility of Engineering to initiate actions that will expedite deliveries. The OPM organization helps identify mitigation plans and monitors their execution with Canadian Manufacturing sites, GO&S and Supply Chain NPI teams.

9.0 Sourcing and Manufacturing Strategy

Prior to Passport 2, the OPE / OPA drives activities to review, validate, influence and achieve agreement on the Program Sourcing Strategy with all stakeholders. The Program Sourcing Strategy form (P&WC 11895) documents agreement on chosen production sources for Key Components. The OPE / OPA ensures that it is flown down to all NPI Sourcing Leads and Managers.

Always ensure compliance with all Contractual obligations associated to the program and to ITC restrictions.

Note: Please refer to OPMWI 3.11.1 Establish Component Strategy for details.

9.1 Advanced Program Sourcing

Per Procurement Process (PPP 06-15), production sources are normally awarded following the receipt and analysis of three production quotes obtained from potential suppliers. This is applicable to all engine parts except for the Key Components that are part of the Program Sourcing Strategy form. In this latter case, considering that the sources are already identified, the OPM Planner must indicate them in the associated Job Tickets to ensure alignment. Please note that even if the source is identified in the Program Sourcing Strategy, Supply Management may elect to still query other sources for benchmarking purpose.

If a sourcing change is requested or required on any Key Component (usually part of the Program Sourcing Strategy form), a formal review is organized by the OPE / OPA and the final agreement is documented in a Component Strategy form (P&WC 10367).

Whenever possible, experimental hardware orders shall be placed at the identified production sources. The objective is to prove the production sources as early as possible to minimize tooling and non-recurring costs, resolve producibility issues and minimize risks at entry into production. In the situation where the lead time is unacceptable to the program, a temporary source can be used if a formal transition strategy is defined and agreed upon by all stakeholders. The Component Strategy (P&WC 10367) form must be used to document this agreement.

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All sourcing changes shall be done prior certification to allow parts substantiation. Otherwise the Work Transition (Resourcing) process will need to be used (refer to MMP 300).

9.2 Dual Sourcing

Prior to Passport 2, the Dual Sourcing Strategy recommendation has been documented in the Program Sourcing Strategy form P&WC 11895. The execution of this strategy is dependent on the decision taken by Senior Management at Passport 1 to fully approve, partially approve or reject the proposal based on the business case results and the associated risks for the program.

At the component level the OPE / OPA, in conjunction with Engineering and Supply Management, ensures progress on the execution of the Dual Sourcing Plan (refer to MMP 348) as agreed at Passport 1 - Permission to Offer. In order to minimize substantiation costs, procurement of second source parts must occur on time to meet the Development Level 2 Plan. In case of a second source change or a new dual sourcing requirement, a formal review is organized by the OPE / OPA and the Component Strategy Sheet is used to document the agreement. Impact on the program business case needs to be assessed if any additional funds are required and a budget owner identified.

At the engine level, the OPE / OPA, in conjunction with the Engine Centre Technical Support Manager, establishes the implementation plan of the second source and ensures progress to meet program requirements dates as per Program Operations Level 2 Plan.

10.0 Budget Management

Prior to Passport 2, the NPI Operations Budget (P&WC 11847) is established by OPM to capture Operations' funds required to industrialize a new program. Program CAR is raised by Strategic Financial Valuation (SFV) and approval must be obtained prior starting detailed design (MMP 515). **Figure 1** below shows the inputs to the NPI Operations Budget including the OPM Program Budget explained in section 10.2.

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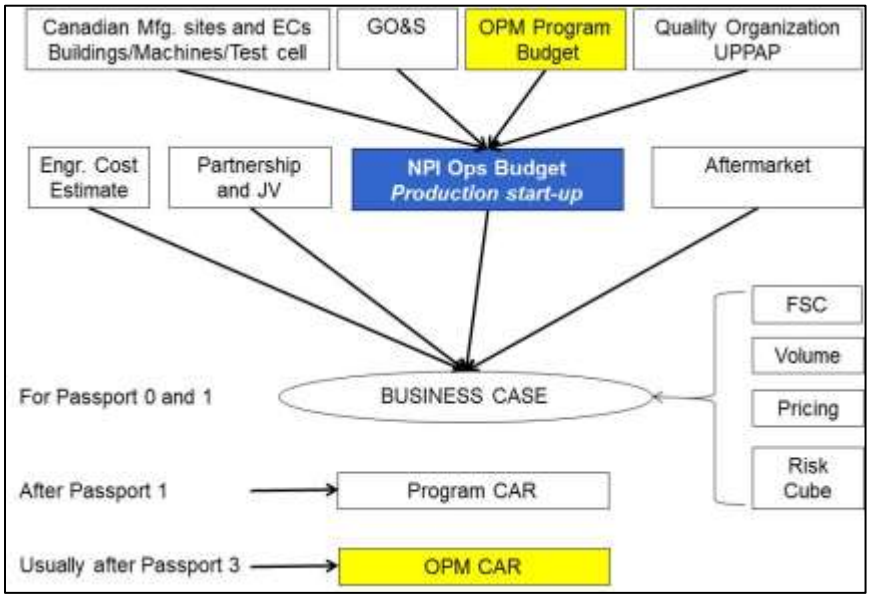

Figure 1 Inputs to NPI Operations Budget

10.1 NPI Operations Budget

The OPE / OPA revises the NPI Operations Budget at every Passport and Investment Audit during the entire NPI phase of the program. In order to prepare for these reviews, update form P&WC 11847 with the actual spend, re-validate all estimates for future activities and add justifications if there are significant differences (+ 10%). A full re-approval signature cycle may be required. The OPE / OPA shall refer to SFV's annual calendar of events in order to plan the updates accordingly.

Note: Please refer to OPMP 3.10 - Industrialization - Pre-Passport 0 up to Passport 2 for more details.

10.2 OPM Program Budget

The OPM Program Budget includes specific capital and managed funds required to industrialize a new or derivative engine. All amounts are captured in tab "OPM CAR & Managed" of form P&WC 11847.

Note: Please refer to OPMWI 3.11.2 Raise OPM Program CAR for more details.

10.2.1 Managed Portion

The managed portion of the OPM Program budget covers the expenses related to the resources required to accomplish the industrialization of a new or derivative program such as Process Planning, 3P event support, Manufacturing Approval, etc. It is indicated in columns "Managed" of the form P&WC 11847 tab "OPM CAR & Managed".

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Once the Program CAR is approved, OPM's Program Managed I.O's can be opened by OPM Finance. A good I.O. structure shall be used in order to facilitate budget monitoring.

10.2.2 Capital Portion

The capital portion of the OPM budget includes production tooling and gauging required by Engine Centres, Canadian Manufacturing facilities and OEM sites to accomplish the industrialization of a new or derivative program. It is indicated in columns "Capital" of the form P&WC 11847 tab "OPM CAR & Managed".

In preparation for Passport 3, the OPE / OPA requests a detailed list of production tools including cost, lead time and phasing of spending. Information is used to raise the OPM CAR shortly after Passport 3 is granted and typically 9 to 12 months prior first production hardware stores date per Program Operations Level 2 Plan.

Upon CAR approval, I.O's can be requested from OPM Finance to launch the tooling procurement process. In case of anticipated budget overspent, a supplement CAR might be required once the original CAR is approved.

10.2.3 OPM Program Budget Monitoring

On a monthly basis, the OPE / OPA monitors the status of expenses using the Consolidated Report. In case the budget is significantly underspent or overspent, corrective actions are required such as raising a supplement CAR. On an annual basis the OPE / OPA provides the latest Program budget estimates to the OPM Budget Coordinator (please refer to OPMWI 2.10.1 OPM Annual Budget Management for more details).

Moving towards Passport 4, the completion of tools procurement cycle is closely monitored with Process Planning or Tooling crib organizations in order to request closure of the I.O's in a timely fashion.

The complete list of NPI Operations Budget I.O's (capital and managed) shall be closed at the aircraft entry into service i.e. when the program becomes Legacy.

11.0 Management of Capital Equipment

In order to industrialize a new Program, Operations sites may need to invest in capital equipment such as production tooling, machines, test cell upgrades or even new buildings. This section explains the role of OPM to ensure that the capital equipment is available as per Program Operations L2 Plan.

Note: Please refer to OPMWI 3.11.3 Manage Capital Equipment for details

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11.1 Canadian Manufacturing

Capital requirements for machines and buildings are managed by each Canadian Manufacturing site. They are responsible to raise their CAR and ensure that all associated actions are taken to support parts manufacturing. The OPE / OPA contacts these organizations on a regular basis to ensure adequate progress on the implementation plan.

For production tooling and gauging, the OPE / OPA provides I.O's to initiate the procurement process at each Canadian Manufacturing site. Tooling delivery is also monitored to ensure Production Readiness of first manufactured part as per Program Operations L2 Plan.

11.2 Engine Centre

Capital requirements for machines, buildings, test cell upgrade and automation are entirely managed by Engine Centre. They are responsible to raise their CAR and ensure that all associated actions are taken to support first engine build. The OPE / OPA communicates with Engine Centre Technical Services on a regular basis to ensure progress on the implementation plan.

For production tooling, the OPE / OPA provides I.O's to initiate the procurement process by Engine Production Definition (EPD). Tooling delivery is also monitored to ensure production readiness for first engine build as per Program Operations L2 Plan.

11.3 OEM Tooling

Procurement of OEM tooling is managed by two organizations. Tools that are common to Engine Centre tooling list are procured by Development Process Planning. Tools that are unique to the OEM are procured by the OPM Planner. The OPE / OPA provides I.O's to both organizations to allow procurement process. Tooling delivery is monitored to ensure availability of the complete set of tools at the OEM for first engine shipment as per Program Operations Level 2 Plan.

11.4 Other Operations Sites

Non-Canadian Manufacturing sites (GO&S) are responsible to raise their CAR and ensure that all actions are taken to complete full implementation before parts production. The OPE / OPA contacts on a regular basis these organizations to ensure progress of their Level 3 plan.

12.0 Program Operations Risks Management

After Passport 2, the OPE / OPA reviews and updates the Program Operations Risks list with all organizations involved. Top operational risks are communicated to Engineering to update the Program Risk cube. It is

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strongly recommended to drive formal reviews with Procurement, Producibility, Engine Centre and Engineering after FER (First Engine Run), FCE (First Customer Engine) and prior FPE (First Production Engine) as new risks may arise with the engine builds.

Note: Please refer to OPMWI 3.10.3 Manage Program Operations Risks for details.

13.0 Production Readiness

Refer to OPMP 3.8 Production Readiness.

14.0 Delivery of Production Engines

Most of the activities required to procure, build and ship the first production engines take place between Passport 3 - Permission to order production hardware and Passport 4 - Permission to deliver serial production. OPM has a key role to coordinate and monitor progress of activities with all organizations involved and address any issues that could jeopardize production engine deliveries.

14.1 IPTs and PMT

As design work progresses toward Passport 3, the OPE / OPA must ensure via the various design IPTs that all issues that can potentially affect production are getting adressed in a timely fashion. Progress of DP closure shall be closely monitored when impacting the production intent BS release for items such as: Engineering Source Approval (ESA), Engineering Red Flag Notice (RFN) / Limit notice (LN) closure and post-cert Change Order (CO) releases.

During the PMT, the OPE / OPA provides a general status on production engine parts and tooling. As mentionned previously in the Development phase section, any concerns on the engine design progress that can result in missing operational and customer commitments in terms of cost, schedule, producibility or quality, shall be highlighted. The OPE / OPA remains responsible to communicate the overall program status provided during the PMT, to Operations' organizations (Canadian Manufacturing sites, Supply Management, Engine Centres, etc.) when deemed necessary.

Note: Refer to the MMP 219, Appendix H to obtain a more detailed list of key Operations deliverables expected at the IPT and PMT levels.

14.2 Production Build Specification

Throughout the Development phase, Engineering releases multiple versions of the engine Build Specification (BS). As we get closer to Passport 3, a configuration freeze representing production intent is required to load SAP and trigger procurement actions.

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Per the Program Operations Level 1 & 2 Plan, release of the production intent BS must be driven to ensure there is sufficient lead time for Supply Chain (including GO&S), Canadian Manufacturing and Engine Centre activities to procure the parts.

The OPE / OPA shall escalate at the PMT any delay foreseen in the release of the production intent BS prior Passport 3. All efforts shall be put forward by Engineering to release the BS on time for Operations. In the situation where the BS release date cannot be met by Engineering, parts lead time analysis is performed by OPM. Multiple recovery actions, supported and agreed upon by Supply Management (GO&S, Canadian Manufacturing and Engine Centre), can be taken depending on the configuration status, such as evaluating part expedite, creating production kit, raising risk orders, etc. (Ref. Appendix A "Guidelines for recovery actions due to Production BS delays"). The proposed Operations recovery plan shall be communicated and agreed upon at the PMT. In all cases, the production intent BS release shall be closely monitored to ensure program and customer commitments are met.

14.3 Load new Build Specification in SAP

The Change Incorporation Analyst (CIA) is responsible to load the production intent BS in SAP. OPM provides material master data for all unique parts while Development Process Planning supplies the AFS structure content. First production engines are expected to require more build time (typically 2 months) due to production manpower learning curve (assemblers, inspectors, ATTS, etc.), audit engine requirements and Manufacturing Approval activities. This initial lead time needs to be agreed between OPM, Engine Centre Logistics and Master Scheduling. As soon as the BS is fully loaded in SAP and Passport 3 granted, the OPE / OPA contacts Master Scheduling to link the engine Master Production Schedule to the BS. From that point forward, production demand is generated at the parts level and any further configuration changes follow the CO process.

When the first production intent BS is loaded in SAP, the packaging section may not be available as it is not fully defined yet. The OPE / OPA shall track the release of the packaging CO via the associated IPT. The objective is to ensure sufficient lead time to procure packaging material, proceed with substantiation and complete the Development PFMEA prior first production engine delivery.

Note: Please refer to OPMWI 3.11.5 Load Production Intent Build Specification in SAP for details

14.4 Production Configuration Management

As soon as the production intent BS is loaded in SAP, OPM needs to keep track of all upcoming COs that will impact production engines.

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- **Mandatory revision letters**

Mandatory revision letters are used when form, fit and function changes affect a part that has not been received yet and for which the supplier or producer is able to incorporate the change with no impact on WIP as the manufacturing is not past the affected operation (refer to OPMP 4.11 for more details).

- **Product improvement COs**

Product improvement COs are driven per the agreed incorporation strategy provided in the COID. In some instances, the PMT may want to accelerate these incorporations to meet a customer commitment, minimize financial impact for P&WC, etc. The OPE / OPA needs to validate with key stakeholders if accelerating the incorporation is achievable. In this case, a Limit Notice (LN) might be used to control inventory of the affected hardware.

- **Mandatory COs**

Mandatory COs are required for all production engines (at P&WC or OEM) prior aircraft entry into service (EIS). There are two types of such COs:

- "Must All" - Change must be incorporated on all engines at P&WC (no engines are delivered at OEM yet).

- "Serial # and Mandatory Retrofit" - Change must be incorporated on all engines at P&WC and OEM.

As soon as a Mandatory Class 1 CO is identified by Engineering to introduce an urgent change on production engines, an Engineering Red Flag (RFN) should be raised against the affected parts to limit inventory and ensure control of configuration.

- **ESA Approval COs**

Status of ESA parts must be monitored as they can prevent engine shipment. Depending on the ESA type (refer to CPW 135), some or all of the following deliverables are required to change the status of an ESA part from "Designated" to "Approved":

1. ESA process submitted to manufacture the part is "Approved" (if applicable);

2. Part has been received with a CofC confirming source and ESA process number (if applicable);

3. Substantiation is completed and DVPM approved.

Following Engine Certification, Engineering completes all DVPMs, close DPs and issue COs to change the status of all ESA parts from "Designated" to "Approved". ESA status are changed using Class 1

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COs for critical parts and via class 2 COs for non-critical parts. From that point on, any new CO affecting an ESA part is tracked carefully to ensure approval prior engine ship.

- **Shipment Restricted COs**

After Engine Certification, any CO with pending substantiation are issued with Shipment Restriction. Tracking of actions required to lift the Shipment Restriction is critical to avoid any impact on the entry into production.

A list of all active Mandatory revision letters, COs, Engineering RFNs and LNs shall be presented at the PMT on a weekly basis to ensure target dates are respected and escalate any slips as per Program's commitments (refer to OPMP 5.10 for CO Tracking List template). This list can also help determine the need for risk hardware and document any upcoming OEUP and / or Return Engine activities (see next section for more details).

Note: Please refer to OPMP 5.10 Change Order Incorporation Management for more details.

14.5 Production Hardware Management

OPM is responsible to ensure that all the necessary hardware is available for first engine build, as per the Program Operations Level 1 & 2 Plan.

Considering that the Integrated Material Flow (IMF) group only tracks part deliveries approximatively two (2) months prior engine build, the OPM Planner provides earlier visibility using the Supply Chain Action Required tool (SCAR). This tool essentially shows if POs are placed on time and if parts deliveries are on track. In case of RFNs, LNs and mandatory COs, the impacted parts are integrated in the tool to ensure full visibility on production configuration.

Any delivery issue that could affect entry into production is escalated and a recovery plan developed in conjunction with Supply Management, Manufacturing sites and Engineering. In the situation where a Mandatory CO release jeopardize hardware availability, OPM can decide to order risk hardware (see **Table 1** below for guidelines). Monitoring of CO release is critical to avoid any hardware delivery in experimental.

In rare situations, when no recovery plan is possible, delaying the first engines deliveries or moving ahead with pre-production engines (refer to section 16 for details) must be discussed at the PMT and decision taken.

Note: Please refer to OPMP 4.11 Production Hardware Management for more details

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Table 1 : Risk hardware criteria and actions

	Prior Engine Certification	Post Engine Certification
Criteria	**1.** Unable to release production intent BS or the follow-on mandatory COs on time considering affected hardware LT **2.** Unique LLT (Finished Parts or details) : Advanced released drawings are available or standard part definition already exists (Industry standard parts, raw material, etc.) **3.** Risk acceptable for Operations	Refer to OPMP 5.10 decision tree
Actions	Order unique LLT Finished Parts, raw material & USM details	Refer to OPMP 5.10 decision tree

14.6 Manufacturing Approval

Manufacturing Approval (MA) is Engine Centre's first article inspection using a production engine to verify that all production processes, documentation, tooling and personnel are capable of producing engines that meet production requirements as per Transport Canada.

In the Program Operations Level 1 & 2 Plan, Manufacturing Approval IPTs start typically 4 months prior first engines build to allow sufficient time to perform all required activities. The OPE / OPA has a key role in coordinating all organizations involved in order to achieve completion of activities per the committed date.

In the situation where the date of completion of the MA activities cannot meet customer requirements, the OPE / OPA shall escalate and seek resolution with the help of the PMT. When no recovery plan is possible, delaying the first engines deliveries or moving ahead with pre-production engines (refer to next section for details) must be discussed at the PMT and a decision taken.

Note: Please refer to OPMP 3.1 Coordinate Manufacturing Approval for details.

15.0 Delivery of Pre-Production Engines

Pre-Production engines (also known as X-engines or Prototype engines) are engines that are built in the production environment but shipped to customer prior Manufacturing Approval (MA) or POA under EASA (for simplicity we will only refer to MA). These engines need to be downgraded prior shipment and they will be upgraded later only once the MA is granted.

The decision to ship Pre-Production engines lies with the PMT. Marketing communicates the Customer's engine requirements. Engineering determines when they can substantiate production hardware and receive Transport

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Canada approval for the new program. OPM determines if production pedigree hardware can be received in time to attain Manufacturing Approval (MA) as per customer requirement. If production hardware is unavailable and the customer requirement cannot be postponed, then engines can only be delivered as Pre-Production engines.

Note 1: Please refer to OPMWI 3.11.6 Manage Pre-Production Engines for details

Note 2: EBU or Nacelle requirements are separate from the engine and therefore have no bearing on the engine's production status.

15.1 Program Downgrade

An engine program that does not have MA requires a program downgrade in order to ship engines. Moreover, if production standard hardware cannot be procured in time for the engines and unsubstantiated hardware can be used, additional downgrade documentation is needed. Refer to **Table 2** for details.

Based on customer engine need dates and the estimated date for MA, the OPE / OPA determines the number of Pre-Production engines that will be required in order to minimize impact on documentation. If there is any uncertainty in the engine serial number range, a larger than needed range should be selected. Once the Manufacturing Approval Engine serial number is confirmed, these documents need only be revised once to pull in. The OPE / OPA is involved in coordinating downgrade activities to ensure shipment of Pre-Production engines per committed date.

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Table 2 : Pre-Production Engines Documentation Requirements

	Downgrade due to Manufacturing Approval not granted	Downgrade due to unsubstantiated hardware
Applicability	Pre-production engines in ALL cases	Pre-production engines with unsubstantiated hardware
Engine Effectivity	All engines as defined in the Downgrade CO, as determined by the OPE/OPA	All engines with unsubstantiated hardware
CR	Single CR required for all downgrade / upgrade COs	
Downgrade CO	Single CO required; supplement if engine range changes	CO required for each design intent (could be multiple)
Upgrade CO	Single CO required; issued when engine range confirmed and MA received	CO required for each design intent (could be multiple); issued when production hardware is available
AQP	Single AQP referencing both downgrade and upgrade COs	
QRA-A	Not required	Required from Engine Centre Quality
RT	Not required	Special testing may be required by Engineering
TATP	Unique TATP required, referencing issued downgrade CO and AQP	Unique TATP required, referencing issued downgrade CO, AQP and QRA-A. Issued RT also referenced if applicable.
OEUP	Required for production standard dataplate for engines to be upgraded at the OEM	Required ideally for each CO, but can be combined if engine effectivity is the same; for engines to be upgraded at the OEM
Build Instructions	Required for installation of production dataplate for engines to be upgraded at the OEM	Required to remove unsubstantiated hardware from engines to be upgraded at the OEM

15.2 Provisioning of Upgrade Hardware

In parallel with the downgrade activities, OPM is responsible for managing hardware to support upgrades of Pre-Production engines to production standard. Depending on the scope of work, these upgrades can be performed either at the OEM or as Return engines at P&WC Engine Centre.

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At a minimum, engine dataplates, rivets and manpower to change the Pre-Production engine dataplates to production are required. Where unsubstantiated hardware is installed, replacement production hardware is required along with manpower for the installation

Note : Please refer to OPMP 6.10 Management of Delivered Engines for more details.

15.3 Program Upgrade

Once MA is granted, the Program Upgrade CO can be issued. Pre-Production engines can now be upgraded to Production status either at the OEM or as Return engines at P&WC Engine Centre. The OPE / OPA is responsible to coordinate the upgrade activities. Unsubstantiated hardware is replaced with Production standard hardware as per the OEUP / Return engine processes OPMP 6.10.

Note: OPM best practice is to only send unique production hardware to the OEM after MA is obtained. If required and agreed upon by the PMT, the hardware upgrade on Pre-Production engines can be initiated prior obtaining MA by issuing Production parts with Form Ones that explain the Prototype status (refer to QMSP 10-02).

When the engine configuration has met the minimum production standard, the engine can receive a new Production engine data plate, logbook and Form One.

In order to be upgraded to Production Status, the engines only need to be upgraded to the Production Standard established at MA. Any product improvements that came after MA are not required in order to upgrade the engine. However they can be done at any time and may be required before the engines can enter service as per Section 14.4. Any upgrades beyond what is required to bring the engines to Production and EIS standards should be reviewed with the PMT and coordinated by the OPE / OPA.

Note: For cases where FAA & Emissions approvals are outstanding, the production data plate will still indicate that FAA and Emissions are "pending" and not "approved". The OPE / OPA determines if the engine upgrades can proceed or wait until FAA & Emissions are "approved". If upgrades proceed, then the data plates will need to be upgraded again prior EIS to show that FAA & Emissions are approved.

16.0 OEM Management

OPM is responsible during Program Industrialization for OEM readiness activities, and in particular for:

- Establishing consignment inventory of replacement material
- Ensuring tooling availability required to perform scheduled and unscheduled maintenance

The timing of these activities is specified in the Program Operations Level 2 Plan. Readiness status of these activities is captured in the Program Production Readiness Dashboard, under the OEM criteria.

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Replacement material and line maintenance tools shall be in place at the OEM as soon as the first production engine is shipped.

Note: Please refer to OPMWI 3.10.4 Establish Program Operations Level 1 and 2 Plan for more details.

16.1 OEM Consignment Inventory

Consignment inventory is PWC-owned engine hardware placed at the OEM's chosen facility under the responsibility of the FSR and is intended for use on production engines only in case of unplanned failure or damage. The objective of consignment is to have readily available at the OEM, parts which, based on past experience, have demonstrated in early stages of programs, quality or technical issues which may require their immediate replacement.

The IPT will together agree on what hardware, and related consumables, is necessary to maintain at the OEM for production start. The activities related to establishing OEM consignment inventory should be initiated when Passport 3 is granted and driven by the longest lead time item to be ordered.

Note : Please refer to OPMWI 3.11.7 Manage OEM Consignment Inventory for more details.

16.2 OEM Tooling

Tooling is also required to be placed at the OEM's facility for possible engine maintenance and repair.

Note: Please refer to OPMWI 3.11.3 Manage Capital Equipment for more details.

16.3 OEM Synchronization

In preparation for production, coordinate with the FSR to receive an OEM production aircraft assembly WIP report on a regular basis; frequency to be established based on volume. This report is useful to understand where the delivered engines are with regard to the OEM schedule.

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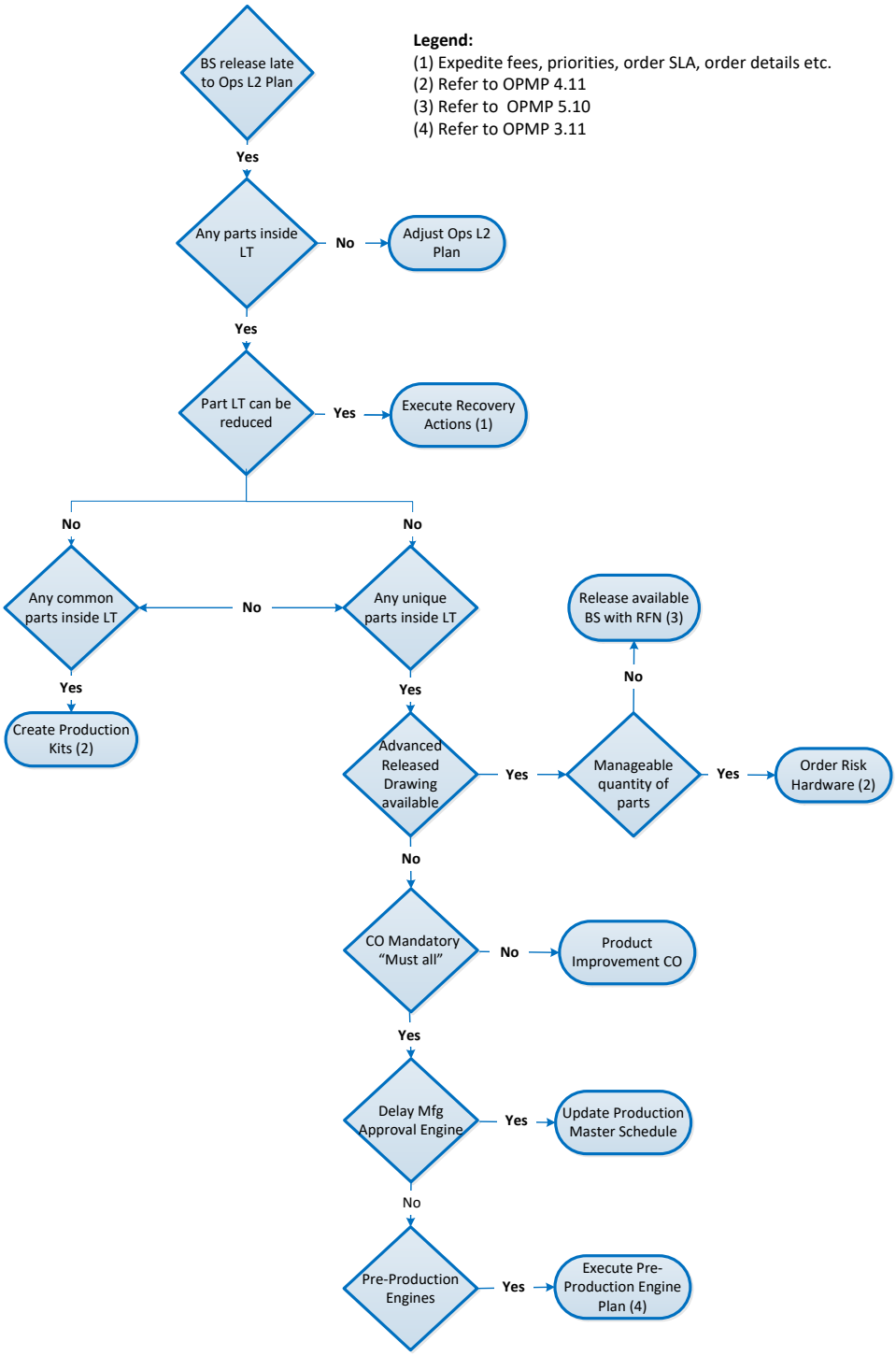
Appendix A: Guidelines for recovery actions due to Production BS delays



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